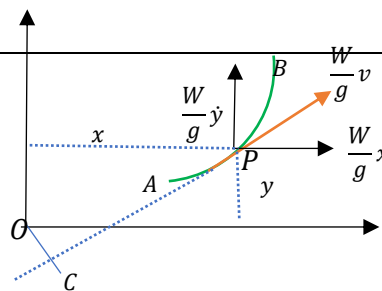


6.6 Moment of Momentum

Consider a particle of weight W moving along a curved path AB in the x - y plane, and let v be the velocity of the particle at any instant of time. Its momentum is defined by $\frac{W}{g}v$. This momentum is a vector tangential to the path in the direction of velocity.



Moment of the momentum of the particle with respect to the origin is the product of momentum and the perpendicular distance OC from the moment centre O to the line of the moment vector.

$$H_o = \left(\frac{W}{g}v\right) \times OC$$

If the velocity components in the x and y directions are $\dot{x}$ and $\dot{y}$, the momentum can also be resolved in to the x - y components as $\frac{W}{g}\dot{x}$ and $\frac{W}{g}\dot{y}$. Therefore, by the Varignon's theorem, the moment of a resultant vector is equal to the algebraic sum of the moments of the components. We can write,

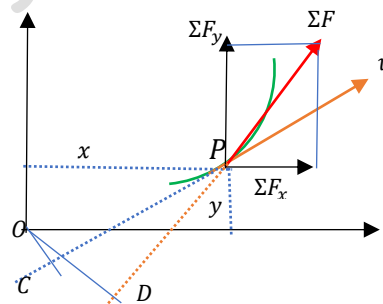
$$H_o = \left(\frac{W}{g}\dot{y}\right) \times x - \left(\frac{W}{g}\dot{x}\right) \times y$$

If ΣF is the net force acting on the body at the given instant, it can also be resolved in to its components ΣF_x and ΣF_y . The moment of ΣF about origin O ,

$$M_o = \Sigma F \times OD$$

Again, by Varignon's theorem,

$$M_o = \Sigma F_y \times x - \Sigma F_x \times y$$



Now, let us consider the equations of curvilinear translation.

$$\frac{W}{g}\ddot{x} = \Sigma F_x \quad \text{and} \quad \frac{W}{g}\ddot{y} = \Sigma F_y$$

Multiplying the first equation by y , second equation by x , and subtracting the first from second

$$\begin{aligned} x \times \frac{W}{g}\ddot{y} - y \times \frac{W}{g}\ddot{x} &= x \times \Sigma F_y - y \times \Sigma F_x \\ x \times \frac{W}{g} \frac{d\dot{y}}{dt} - y \times \frac{W}{g} \frac{d\dot{x}}{dt} &= x \times \Sigma F_y - y \times \Sigma F_x \\ \frac{d}{dt} \left(x \times \frac{W}{g}\dot{y} \right) - \frac{d}{dt} \left(y \times \frac{W}{g}\dot{x} \right) &= x \times \Sigma F_y - y \times \Sigma F_x \\ \frac{d}{dt} \left[\left(x \times \frac{W}{g}\dot{y} \right) - \left(y \times \frac{W}{g}\dot{x} \right) \right] &= x \times \Sigma F_y - y \times \Sigma F_x \\ \frac{d}{dt} H_o &= M_o \end{aligned}$$

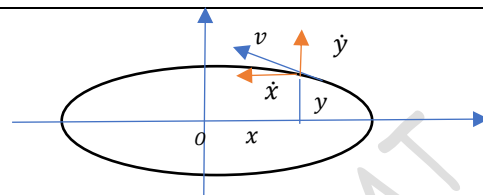
The equation states that the rate of change of moment of momentum of a particle with respect to any point in its plane of motion is equal to the moment of the resultant force with respect to the same point. If the moment of the external forces acting on the body is zero, then its moment of momentum remains constant.

Example 1

The motion of a particle of mass m in the $x - y$ plane is defined by the equations $x = a \cos pt$, and $y = b \sin pt$, where a, b, p are constants. Calculate the moment of momentum of the particle with respect to the origin.

Given $x = a \cos pt$ and $y = b \sin pt$. This is the equation of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

$$\dot{x} = -ap \sin pt \quad \text{and} \quad \dot{y} = bp \cos pt$$

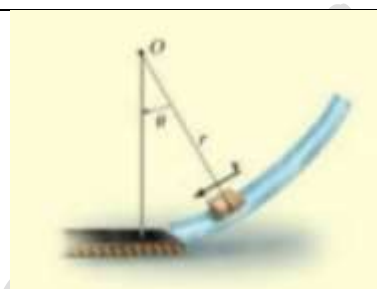


Moment of momentum of the particle about $O = (x \cdot m \dot{y} + y \cdot m \dot{x})$ (both are positive as they are c.c.w)

$$H_o = m(x \cdot \dot{y} + y \cdot \dot{x}) = m(a \cos pt \cdot bp \cos pt + b \sin pt \cdot ap \sin pt) = mpab(\sin^2 pt + \cos^2 pt) = mabp$$

Example 2

The box shown in figure has a mass m and is travelling down the smooth circular ramp such that when it is at the angle θ , it has a speed v . Compute its moment of momentum about point O at this instant and its tangential acceleration.

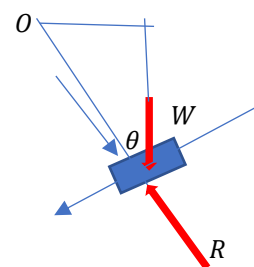


Since the velocity v is tangent to the path, its moment of momentum

$$H_o = mv \times r = mvr$$

We know that,

$$\frac{d}{dt} H_o = M_o$$

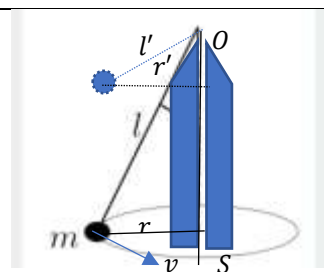


$$\frac{d}{dt} (mvr) = W \sin \theta \times r$$

$$mr \frac{dv}{dt} = mgr \sin \theta \gggggg \text{ tangential acceleration, } \frac{dv}{dt} = g \sin \theta$$

Example 3

A conical pendulum of length $l = 50 \text{ cm}$ rotates with constant speed v in a horizontal circular path of radius $r = 25 \text{ cm}$ as shown. How much the string must be pulled through the pedestal to double the speed of the ball?



The pendulum rotates about the axis OS . The forces acting on the system are weight W and tension T . The weight W is parallel to the axis and tension T is intersecting with the axis. Hence, the moments of these forces about the axis is zero. Therefore, the moment of momentum is constant

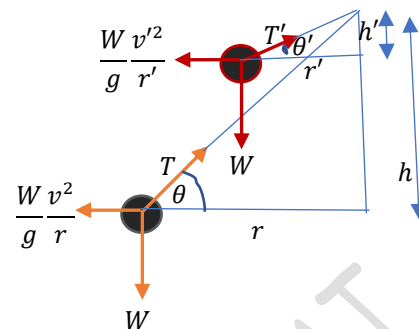
$$v' = 2v$$

$$mvr = mv'r'$$

$$r' = \frac{r}{2} = 12.5 \text{ cm}$$

$$h = \sqrt{50^2 - 25^2} = 43.3 \text{ cm}$$

Consider the free body diagrams of the bob at the two positions. Since, the conical pendulum rotates with constant speed, there is no tangential acceleration. The normal acceleration is directed towards the centre of the circular path. Incorporating the normal inertia force in to the system, we can write the algebraic sum of all the forces equal to zero as per the D'Alembert's principle.



$$\frac{W v^2}{g r} - T \cos \theta = 0 \quad \text{and} \quad T \sin \theta - W = 0$$

From these two equations, we can write

$$\tan \theta = \frac{r g}{v^2}$$

$$\frac{W v'^2}{g r'} - T' \cos \theta' = 0 \quad \text{and} \quad T' \sin \theta' - W = 0'$$

From these two equations, we can write

$$\tan \theta' = \frac{r' g}{v'^2}$$

$$\tan \theta' = \frac{r' g}{v'^2} = \frac{\left(\frac{r}{2}\right) g}{(2v)^2} = \frac{r g}{8 v^2} = \frac{\tan \theta}{8}$$

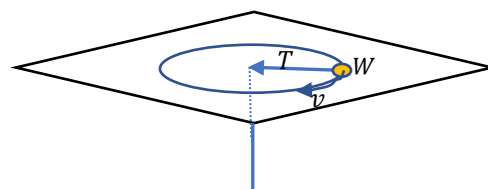
$$\text{i.e.,} \quad \frac{h'}{r'} = \frac{h}{8r} \gggg h' = \frac{43.3 \times 12.5}{8 \times 25} = 2.7 \text{ cm}$$

$$l' = \sqrt{h'^2 + r'^2} = \sqrt{2.7^2 + 12.5^2} = 12.8 \text{ cm}$$

The amount of cord which need to be pulled down = $l - l' = 50 - 12.8 = 37.2 \text{ cm}$

#Exercise 1

A small ball of weight W , attached to the end of a string, is supported by a smooth horizontal plane and travels with uniform speed v in a circular path of radius r . By pulling the string at the lower end, the radius of the path is reduced to $\frac{r}{2}$. Determine the new velocity of the ball and the tension in the string.



#Exercise 2

The motion of a particle of mass m in the $x - y$ plane is defined by the equations $x = a \cos pt$, $y = b \sin 2pt$ where a, b, p are constants. Calculate the moment of momentum of the particle with respect to the origin.

Prof. Biju N., School of Engineering, CUSAT